

Total Marks – 100**Attempt Questions 1-4****All questions are of equal value****Question 1** (25 marks) Use a SEPARATE writing booklet.**Marks**

(a) Evaluate $\int_0^3 \frac{2dx}{\sqrt{x^2 + 16}}$. 2

(b) Find $\int \frac{\sin x}{\cos^3 x} dx$. 2

(c) Find $\int \sin^3 x dx$. 3

(d) (i) Prove $2 \cos A \sin B = \sin(A + B) - \sin(A - B)$. 1

(ii) Hence, evaluate $\int_0^{\frac{\pi}{12}} \cos 4x \sin 2x dx$. 2

(e) Find, by completing the square $\int \frac{1}{x^2 + 6x + 10} dx$. 2

(f) (i) Find real numbers A, B, C such that

$$\frac{3x^2 - x + 8}{(1-x)(x^2 + 1)} = \frac{A}{1-x} + \frac{Bx + C}{x^2 + 1}$$
3

(ii) Hence find $\int \frac{3x^2 - x + 8}{(1-x)(x^2 + 1)} dx$. 2

(g) Evaluate, $\int_0^{\frac{1}{2}} \tan^{-1} x dx$ using integration by parts. 4

(h) Use the substitution $t = \tan \frac{x}{2}$ to find $\int_0^{\frac{\pi}{2}} \frac{1}{5 + 3 \cos x - 4 \sin x} dx$. 4

Question 2 (25 marks) Use a SEPARATE writing booklet.**Marks**

- (a) If $z = 3 - 4i$ and $w = 2 + 5i$,
Express in the form $(x + iy)$ where x and y are real.
- | | | |
|-------|-----------------|---|
| (i) | zw | 1 |
| (ii) | $\overline{zw}$ | 1 |
| (iii) | $\sqrt{z}$ | 2 |
| (iv) | $\frac{z}{w}$ | 2 |
- (b) (i) Express in modulus argument form, $\sqrt{3} + i$. 1
- (ii) Hence evaluate $(\sqrt{3} + i)^5$ in the form $x + iy$. 3
- (c) Graph the region in the Argand diagram which satisfies $|z + \bar{z}| \leq 2$ and $|z - i| \leq 2$ simultaneously. 3
- (d) Let ω be one of the non real cube roots of 1.
- | | | |
|------|--|---|
| (i) | Show $1 + \omega + \omega^2 = 0$. | 1 |
| (ii) | Hence find the value of $(2 - \omega)(2 - \omega^2)(2 - \omega^4)(2 - \omega^5)$. | 2 |
- (e) By applying de Moivre's Theorem and by also expanding $(\cos \theta + i \sin \theta)^5$ express $\cos 5\theta$ as a polynomial in $\cos \theta$. 4
- (f) The origin O and points A, B, C representing the numbers z , $\frac{1}{z}$ and $z + \frac{1}{z}$ respectively are joined to form a quadrilateral. Write down the condition or conditions for z so that the quadrilateral OABC will be
- | | | |
|------|-----------|---|
| (i) | a rhombus | 1 |
| (ii) | a square | 1 |
- (g) (i) Find all solutions of $z^6 = -1$. 2
- (ii) Plot the solutions on the Argand diagram, indicating z_1 . 1

Question 3 (25 marks) Use a SEPARATE writing booklet.**Marks**

- (a) Draw neat sketches of the each following, clearly indicating any intercepts, asymptotes, endpoints, turning points and discontinuities. Make each separate one-third page.
- | | | |
|--------|---|----------|
| (i) | $f(x) = 4 - x^2$ | 1 |
| (ii) | $y = \frac{1}{f(x)}$ | 2 |
| (iii) | $y = \sqrt{f(x)}$ | 2 |
| (iv) | $y^2 = f(x)$ | 2 |
| (v) | $y = f(x) $ | 2 |
| (vi) | $y = [f(x)]^2$ | 2 |
| (vii) | $ y = f(x)$ | 2 |
| (viii) | $y = f(x)$ | 2 |
| (ix) | $y = \log_e f(x)$ | 2 |
| (x) | $y = f(e^x)$ | 2 |
| (b) | Find the gradient of the tangent to the curve, $x^2 - xy + y^3 = 5$ at the point $(1, -2)$. | 2 |
| (c) | (i) State the domain and range of $y = \sin^{-1}(e^x)$. | 2 |
| | (ii) Sketch the graph of $y = \sin^{-1}(e^x)$ showing clearly the co-ordinates of any endpoints and the equation of any asymptotes. | 2 |

Question 4 (25 marks) Use a SEPARATE writing booklet. **Marks**

- (a) Given that $3 - i$ is a zero of the function $P(x) = x^3 + ax^2 + bx - 10$, where a and b are real, find the other zeros and the values of a and b . **4**

- (b) (i) Find all the roots of the equation $x^4 + x^3 - 3x^2 - 5x - 2 = 0$, given there is a root of multiplicity 3. **4**

- (ii) Sketch $y = x^4 + x^3 - 3x^2 - 5x - 2$, showing all intercepts. **2**

- (c) The roots of a certain cubic equation are α , β and γ . Given the following **2**

$$\alpha + \beta + \gamma = -3$$

$$\alpha^2 + \beta^2 + \gamma^2 = 29$$

$$\alpha\beta\gamma = -6$$

Form the cubic equation whose roots are α , β and γ .

- (d) Factorise $P(x) = x^4 - 5x^3 + 4x^2 + 2x - 8$ over

- (i) $\mathbb{R}$ (all real numbers) **3**

- (ii) $\mathbb{C}$ (all complex numbers) **2**

- (e) The equation $x^3 + 2x^2 + bx - 16 = 0$ has roots α , β , γ such that. $\alpha\beta = 4$

- (i) Show that $b = -20$. **2**

- (ii) Find the cubic equation with roots given by α^2 , β^2 and γ^2 . **3**

- (iii) Find the value of $\alpha^3 + \beta^3 + \gamma^3$. **3**

End of paper

Question 1

$$\begin{aligned} \text{a) } \int_0^3 \frac{2}{\sqrt{x^2+16}} dx &= 2 \ln [x + \sqrt{x^2+16}]_0^3 \\ &= 2 \ln [3 + \sqrt{3^2+16}] - 2 \ln [0 + \sqrt{0^2+16}] \\ &= 2 \ln 8 - 2 \ln 4 \\ &= 2 \ln \left(\frac{8}{4}\right) \\ &= 2 \ln 2 \end{aligned}$$

$$\begin{aligned} \text{b) } \int \frac{\sin x}{\cos^3 x} dx &= - \int \cos^{-3} x (-\sin x) dx \\ &= \frac{1}{2} \cos^{-2} x + C \\ &= \frac{1}{2 \cos^2 x} + C \\ &= \frac{1}{2} \sec^2 x + C \end{aligned}$$

$$\begin{aligned} \text{c) } \int \sin^3 x dx &= \int \sin^2 x \sin x dx \\ &= \int (1 - \cos^2 x) \sin x dx \\ &= \int (\sin x - \cos^2 x \sin x) dx \\ &= -\cos x + \frac{1}{3} \cos^3 x + C \\ &= \frac{1}{3} \cos^3 x - \cos x + C \end{aligned}$$

$$\begin{aligned} \text{d) (i) } 2 \cos A \sin B &= \sin(A+B) - \sin(A-B) \\ \text{L.H.S.} &= \sin A \cos B + \cos A \sin B - (\sin A \cos B \\ &\quad - \cos A \sin B) \\ &= 2 \cos A \sin B \\ &= \text{R.H.S.} \\ \therefore 2 \cos A \sin B &= \sin(A+B) - \sin(A-B) \end{aligned}$$

$$\begin{aligned}
 \text{(ii)} \quad \int_0^{\frac{\pi}{12}} \cos 4x \sin 2x dx &= \frac{1}{2} \int_0^{\frac{\pi}{12}} (\sin 6x - \sin 2x) dx \\
 &= \frac{1}{2} \left[-\frac{1}{6} \cos 6x + \frac{1}{2} \cos 2x \right]_0^{\frac{\pi}{12}} \\
 &= \frac{1}{2} \left\{ \left[-\frac{1}{6} \cos \frac{\pi}{2} + \frac{1}{2} \cos \frac{\pi}{6} \right] - \left[-\frac{1}{6} \cos 0 + \frac{1}{2} \cos 0 \right] \right\} \\
 &= \frac{1}{2} \left\{ \frac{\sqrt{3}}{4} - \left[-\frac{1}{6} + \frac{1}{2} \right] \right\} \\
 &= \frac{\sqrt{3}}{8} - \frac{1}{6}
 \end{aligned}$$

$$\begin{aligned}
 \text{e)} \quad \int \frac{1}{x^2 + 6x + 10} dx &= \int \frac{dx}{(x+3)^2 + 1} \\
 &= \tan^{-1}(x+3) + C
 \end{aligned}$$

$$\begin{aligned}
 \text{f)} \quad \frac{3x^2 - x + 8}{(1-x)(x^2+1)} &= \frac{A}{1-x} + \frac{Bx+C}{x^2+1} \\
 \therefore 3x^2 - x + 8 &= A(x^2+1) + (Bx+C)(1-x)
 \end{aligned}$$

$$\text{Put } x=1 \quad 10 = 2A$$

$$A = 5$$

Equating coefficients of x^2

$$3 = A - B$$

$$3 = 5 - B$$

$$B = 2$$

Equating constants

$$8 = A + C$$

$$8 = 5 + C$$

$$C = 3$$

$$\therefore A=5, B=2, C=3$$

$$\begin{aligned}
 \int \frac{3x^2 - x + 8}{(1-x)(x^2+1)} dx &= \int \left(\frac{5}{(1-x)} + \frac{2x+3}{x^2+1} \right) dx \\
 &= \int \left(\frac{5}{1-x} + \frac{2x}{x^2+1} + \frac{3}{x^2+1} \right) dx \\
 &= -5 \ln |1-x| + \ln(x^2+1) + 3 \tan^{-1} x + C \\
 &= \ln \left[\frac{(x^2+1)}{|1-x|^5} \right] + 3 \tan^{-1} x + C
 \end{aligned}$$

$$g) \int_0^{\frac{1}{2}} \tan^{-1} x \, dx$$

$$\text{Let } u = \tan^{-1} x \quad \frac{dv}{dx} = 1$$

$$\frac{du}{dx} = \frac{1}{1+x^2} \quad v = x$$

$$\int u \frac{dv}{dx} dx = uv - \int v \frac{du}{dx} dx$$

$$\int_0^{\frac{1}{2}} \tan^{-1} x \, dx = \left[x \tan^{-1} x \right]_0^{\frac{1}{2}} - \int_0^{\frac{1}{2}} \frac{x}{1+x^2} dx$$

$$= \left[x \tan^{-1} x \right]_0^{\frac{1}{2}} - \frac{1}{2} \left[\ln(x^2+1) \right]_0^{\frac{1}{2}}$$

$$= \frac{1}{2} \tan^{-1} \frac{1}{2} - \frac{1}{2} \left(\ln \frac{5}{4} - \ln 1 \right)$$

$$= \frac{1}{2} \tan^{-1} \frac{1}{2} - \frac{1}{2} \ln \frac{5}{4}$$

$$h) \int_0^{\frac{\pi}{2}} \frac{1}{5+3\cos x - 4\sin x} dx$$

$$= \int_0^1 \frac{1}{5+3\left(\frac{1-t^2}{1+t^2}\right) - 4\left(\frac{2t}{1+t^2}\right)} \times \frac{2}{1+t^2} dt$$

$$= \int_0^1 \frac{2 dt}{5(1+t^2) + 3(1-t^2) - 4(2t)}$$

$$= \int_0^1 \frac{2 dt}{5 + 5t^2 + 3 - 3t^2 - 8t}$$

$$= \int_0^1 \frac{2 dt}{2(t^2 - 4t + 4)}$$

$$= \int_0^1 \frac{dt}{(t-2)^2}$$

$$= \left[\frac{-1}{t-2} \right]_0^1$$

$$= \left[\frac{-1}{1-2} \right] - \left[\frac{-1}{0-2} \right]$$

$$= 1 - \frac{1}{2}$$

$$= \frac{1}{2}$$

$$t = \tan \frac{x}{2}$$

$$\cos x = \frac{1-t^2}{1+t^2}$$

$$\sin x = \frac{2t}{1+t^2}$$

$$\frac{dt}{dx} = \frac{1}{2} \sec^2 \frac{x}{2}$$

$$= \frac{1}{2} (1+t^2)$$

$$dx = \frac{2}{1+t^2} dt$$

$$\text{when } x = 0, t = \tan \frac{x}{2} = 0$$

$$\text{when } x = \frac{\pi}{2}, t = \tan \frac{\pi}{2} = 1$$

Question 2

a) $z = 3 - 4i$, $w = 2 + 5i$

$$(i) zw = (3 - 4i)(2 + 5i) \\ = 26 + 7i$$

$$(ii) \overline{zw} = 26 - 7i$$

$$(iii) x + iy = \sqrt{z} \\ (x + iy)^2 = 3 - 4i \\ x^2 - y^2 = 3 \\ 2xy = -4 \\ xy = -2 \\ x = \pm 2, y = \mp 1$$

$$\therefore \sqrt{z} = 2 - i, -2 + i$$

$$(iv) \frac{z}{w} = \frac{3 - 4i}{2 + 5i} \\ = \frac{3 - 4i}{2 + 5i} \times \frac{2 - 5i}{2 - 5i} \\ = \frac{-14 - 23i}{29}$$

b (i) $z = \sqrt{3} + i$

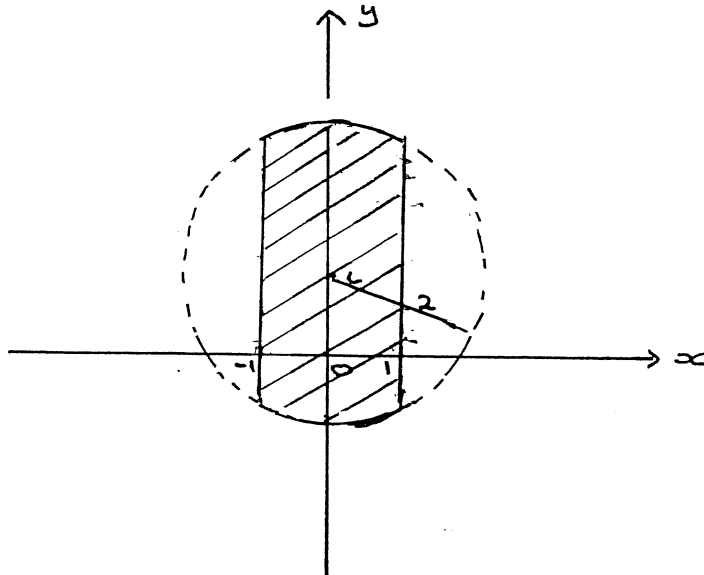
$$|z| = \sqrt{(\sqrt{3})^2 + 1^2} \\ = 2$$

$$\arg z = \tan^{-1}\left(\frac{1}{\sqrt{3}}\right) \quad -\pi < \theta \leq \pi \\ = \frac{\pi}{6}$$

$$\therefore \sqrt{3} + i = 2 \cos \frac{\pi}{6}$$

$$(ii) (\sqrt{3} + i)^5 = 2^5 \cos \frac{5\pi}{6} \\ = 32 \left(\cos \frac{5\pi}{6} + i \sin \frac{5\pi}{6} \right) \\ = 32 \left(-\frac{\sqrt{3}}{2} + i \frac{1}{2} \right) \\ = -16\sqrt{3} + 16i$$

c)



$$|z + \bar{z}| \leq 2$$

$$|x + iy + x - iy| \leq 2$$

$$|2x| \leq 2$$

$$|x| \leq 1$$

d) i) $\omega^3 = 1$

$$\omega^3 - 1 = 0$$

$$(\omega - 1)(\omega^2 + \omega + 1) = 0$$

Since $\omega - 1 \neq 0$, $\omega \neq 1$ ω is not rational

$$\omega^2 + \omega + 1 = 0$$

ii) $(2 - \omega)(2 - \omega^2)(2 - \omega^4)(2 - \omega^5)$ $\omega^3 = 1$

$$= (2 - \omega)(2 - \omega^2)(2 - \omega)(2 - \omega^2)$$

$$= [(2 - \omega)(2 - \omega^2)]^2$$

$$= (4 - 2\omega - 2\omega^2 + \omega^3)^2 \quad \omega^3 = 1$$

$$= [5 - 2(\omega + \omega^2)]^2 \quad \omega^2 + \omega + 1 = 0$$

$$\omega^2 + \omega = -1$$

$$= (5 - 2(-1))^2$$

$$= 7^2$$

$$= 49$$

e) $(\cos \theta + i \sin \theta)^5 = \cos^5 \theta + 5 \cos^4 \theta (i \sin \theta) + 10 \cos^3 \theta (i^2 \sin^2 \theta) + 10 \cos^2 \theta (i^3 \sin^3 \theta) + 5 \cos \theta (i^4 \sin^4 \theta) + (i^5 \sin^5 \theta)$

$$\cos 5\theta + i \sin 5\theta = \cos^5 \theta + 5 \cos^4 \theta (i \sin \theta) - 10 \sin^2 \theta \cos^3 \theta - 10 \cos^2 \theta (i \sin \theta) + 5 \cos \theta \sin^4 \theta + i \sin^5 \theta$$

Equating real parts

$$\cos 5\theta = \cos^5 \theta - 10 \sin^2 \theta \cos^3 \theta + 5 \cos \theta \sin^4 \theta$$

$$\begin{aligned}
 \therefore \cos 5\theta &= \cos^5 \theta - 10 \sin^2 \theta \cos^3 \theta + 5 \cos \theta \sin^4 \theta \\
 &= \cos^5 \theta - 10(1 - \cos^2 \theta) \cos^3 \theta + 5 \cos \theta (1 - \cos^2 \theta)^2 \\
 &= \cos^5 \theta - 10 \cos^3 \theta + 10 \cos^5 \theta + 5 \cos \theta (1 - 2 \cos^2 \theta + \cos^4 \theta) \\
 &= \cos^5 \theta - 10 \cos^3 \theta + 10 \cos^5 \theta + 5 \cos \theta - 10 \cos^3 \theta + 5 \cos^5 \theta \\
 &= 16 \cos^5 \theta - 20 \cos^3 \theta + 5 \cos \theta
 \end{aligned}$$

f) (i) $|z| = 1$

(ii) $|z| = 1$ and $\arg z = \frac{\pi}{2}$

$$|z| = 1, \quad |z + \frac{1}{z}| = |z - \frac{1}{z}|$$

$$\text{or } \arg\left(\frac{1}{z}\right) - \arg z = \frac{\pi}{2} \quad \text{or } z^2 + \frac{1}{z^2} = 0$$

g) $z^6 = -1$

$$\text{Let } r(\cos \theta + i \sin \theta) = \sqrt[6]{-1}$$

$$r^6(\cos 6\theta + i \sin 6\theta) = -1$$

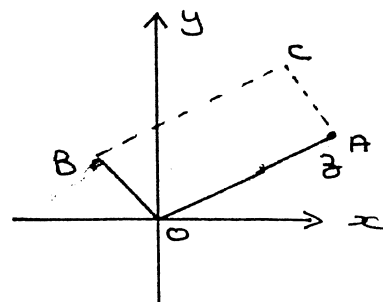
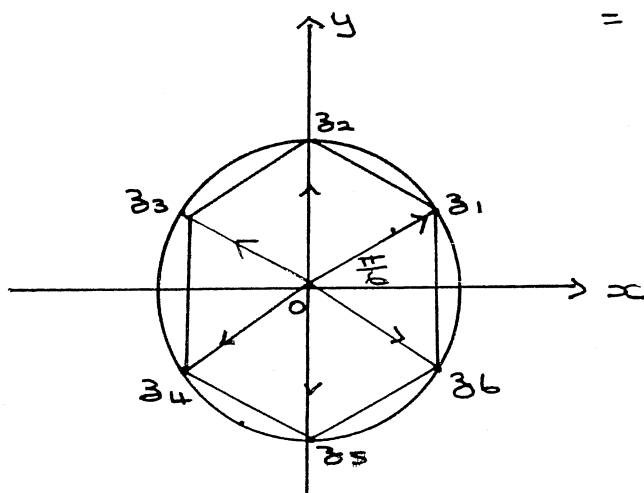
$$\cos 6\theta = -1$$

$$6\theta = (2k+1)\pi$$

$$\theta = \frac{(2k+1)\pi}{6} \quad k = 0, 1, 2, 3, 4, 5$$

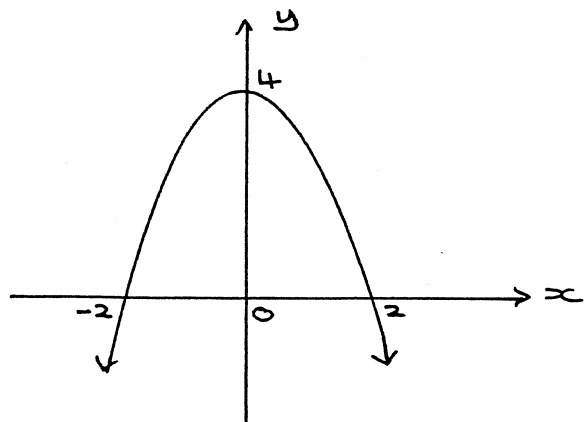
$$= \frac{\pi}{6}, \frac{\pi}{2}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{3\pi}{2}, \frac{11\pi}{6}$$

$$\frac{9\pi}{6} = \frac{3\pi}{2}, \quad \frac{11\pi}{6} = \frac{11\pi}{6}$$

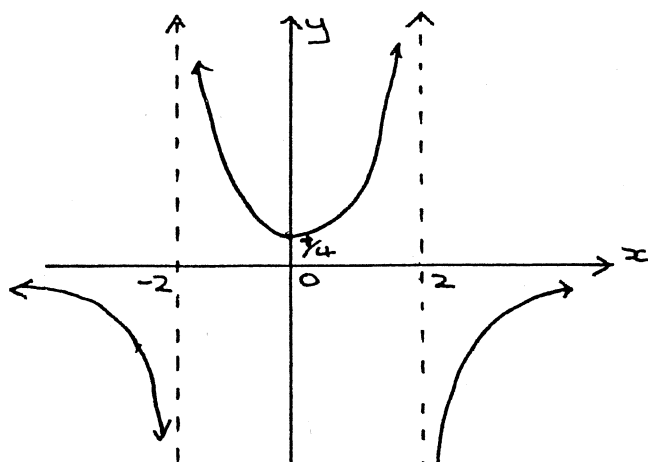


Question 3

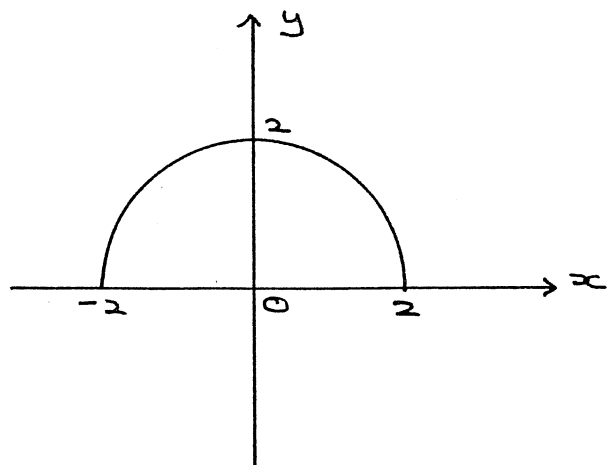
a) $y = 4 - x^2$



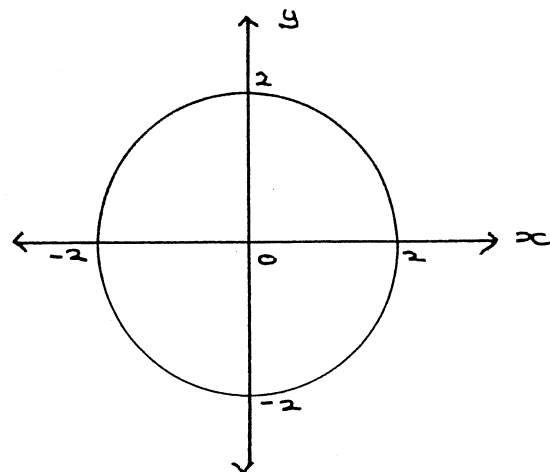
ii) $y = \frac{1}{f(x)}$ $x \neq \pm 2$



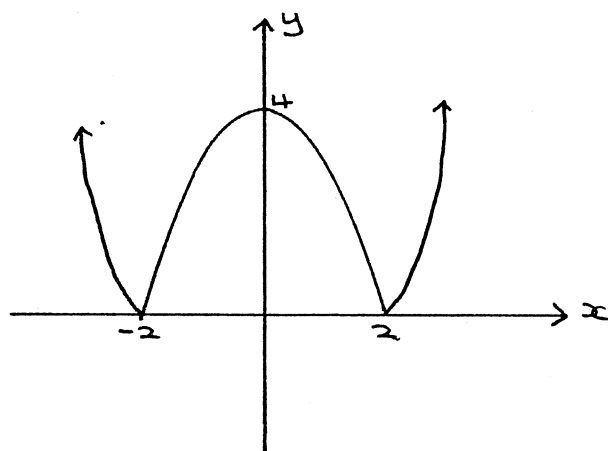
iii) $y = \sqrt{f(x)}$ $4 - x^2 \geq 0$
 $-2 \leq x \leq 2$



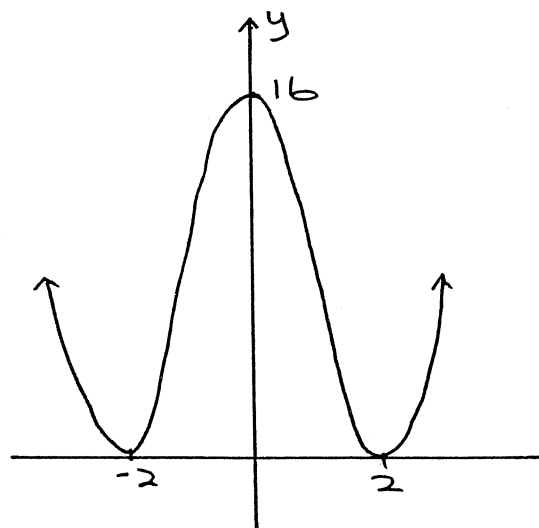
iv) $y^2 = f(x)$
 $y = \pm \sqrt{f(x)}$



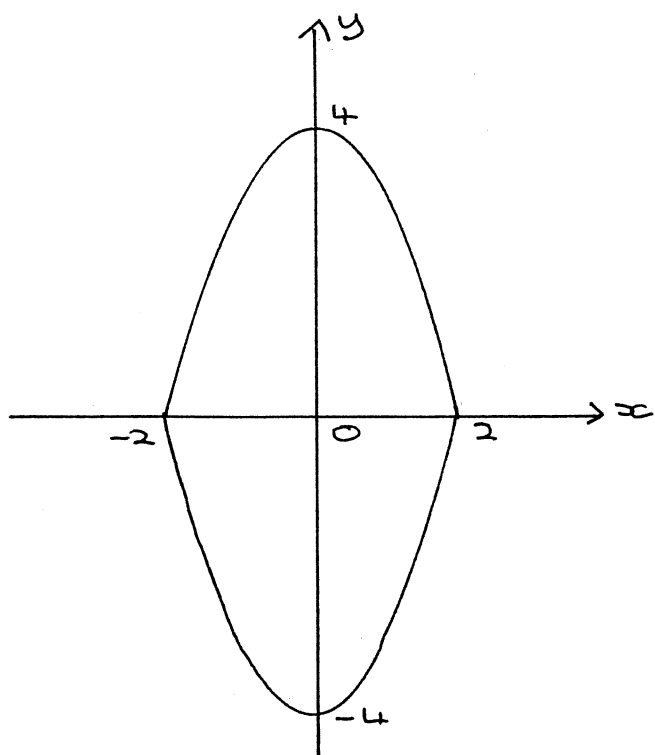
v) $y = |f(x)|$



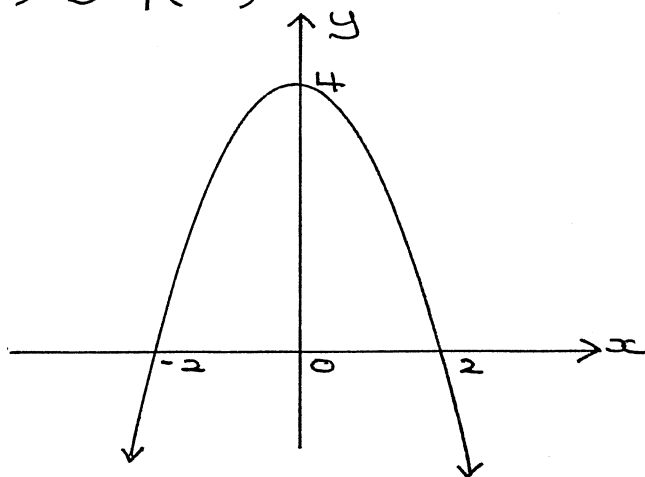
vi) $y = [f(x)]^2$



vii) $|y| = f(x)$



viii) $y = f(|x|)$



ix) $y = \log_e f(x)$

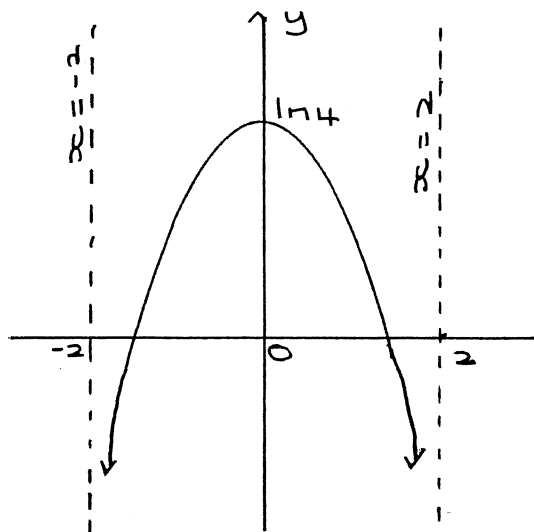
$$f(x) > 0$$

$$x\text{-intercept} \Rightarrow y = 0$$

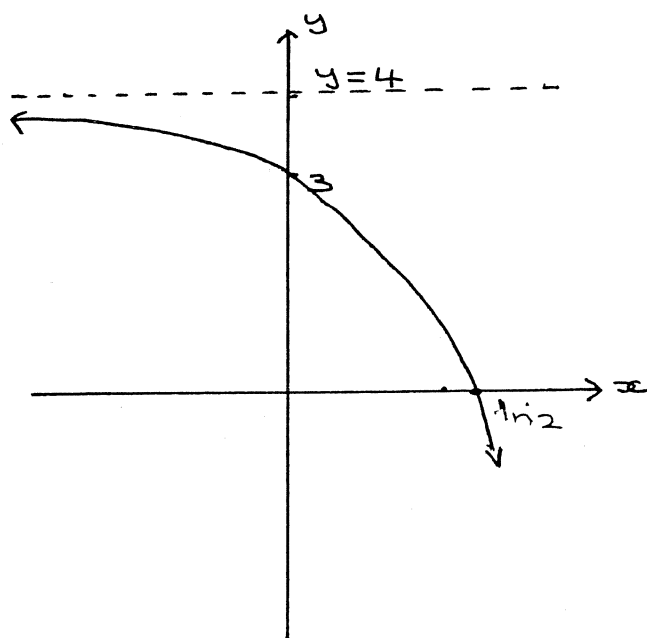
$$\therefore 4 - x^2 = 1$$

$$x^2 = 3$$

$$x = \pm\sqrt{3}$$



$$\begin{aligned} x) \quad y &= f(e^x) \\ &= 4 - (e^x)^2 \\ &= 4 - e^{2x} \end{aligned}$$



$$x\text{-intercept} \Rightarrow y = 0$$

$$4 - e^{2x} = 0$$

$$e^{2x} = 4$$

$$2x = \ln 4$$

$$x = \ln 2$$

b) $x^2 - xy + y^3 = 5$
 $2x - (y + x \frac{dy}{dx}) + 3y^2 \frac{dy}{dx} = 0$
 $2x - y - x \frac{dy}{dx} + 3y^2 \frac{dy}{dx} = 0$

$$\frac{dy}{dx}(3y^2 - x) = y - 2x$$

$$\frac{dy}{dx} = \frac{y - 2x}{3y^2 - x} \quad \text{at } x=1, y=-2$$

$$= \frac{-2 - 2}{12 - 1}$$

$$= \frac{-4}{11}$$

c) i) $y = \sin^{-1}(e^x)$

domain $-1 \leq e^x \leq 1$

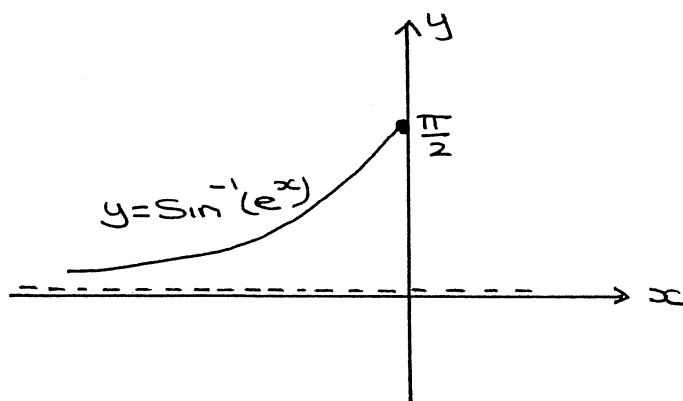
$$0 \leq e^x \leq 1$$

$$x \leq 0$$

Range $0 < e^x \leq 1$

$$0 < \sin^{-1} e^x \leq \frac{\pi}{2}$$

$$0 < y \leq \frac{\pi}{2}$$



Question 4

- a) If $3-i$ is a root then $3+i$ is also a root by the conjugate root theorem

$$P(x) = x^3 + ax^2 + bx - 10$$

$$\alpha\beta\gamma = 10$$

$$(3-i)(3+i)\gamma = 10$$

$$10\gamma = 10$$

$$\gamma = 1$$

$\therefore$ The roots are $3-i$, $3+i$, 1

$$\sum \alpha = -a$$

$$3-i+3+i+1 = -a$$

$$a = -7$$

$$\sum \alpha\beta = b$$

$$3-i+3+i+(3-i)(3+i) = b$$

$$b = 16$$

- b) Let $P(x) = x^4 + x^3 - 3x^2 - 5x - 2$
 $P'(x) = 4x^3 + 3x^2 - 6x - 5$
 $P''(x) = 12x^2 + 6x - 6$

By the Multiple Root Theorem

$$P''(x) = P'(x) = P(x) = 0$$

$$12x^2 + 6x - 6 = 0$$

$$2x^2 + x - 1 = 0$$

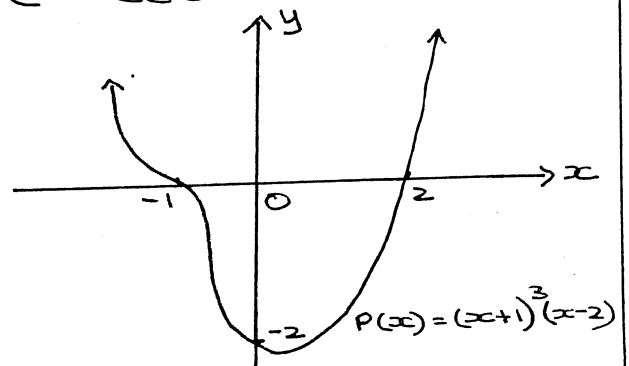
$$(2x-1)(x+1) = 0$$

$$x = \frac{1}{2}, -1$$

$$\therefore P''(1) = P'(1) = P(1) = 0$$

$\therefore x = -1$ is a triple root

$$\therefore P(x) = (x+1)^3(x-2)$$



$$\begin{aligned} \text{c) } \alpha + \beta + \gamma &= -3 \quad \text{--- ①} \\ \alpha^2 + \beta^2 + \gamma^2 &= 29 \quad \text{--- ②} \\ \alpha\beta\gamma &= -6 \quad \text{--- ③} \end{aligned}$$

$$\alpha^2 + \beta^2 + \gamma^2 = (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \alpha\gamma + \beta\gamma)$$

$$29 = (-3)^2 - 2(\alpha\beta + \alpha\gamma + \beta\gamma)$$

$$\alpha\beta + \alpha\gamma + \beta\gamma = -10 \quad \text{--- ④}$$

$\therefore x^3 + 3x^2 - 10x + 6 = 0$ is the required polynomial

$$\text{d) i) } P(x) = x^4 - 5x^3 + 4x^2 + 2x - 8$$

$$P(-1) = 1 + 5 + 4 - 2 - 8 = 0$$

$\therefore (x+1)$ is a factor

By synthetic division

$$\begin{array}{r|rrrrrr} -1 & 1 & -5 & 4 & 2 & -8 \\ & 0 & -1 & 6 & -10 & 8 \\ \hline & 1 & -6 & 10 & -8 & 0 \end{array}$$

$$\therefore P(x) = x^4 - 5x^3 + 4x^2 + 2x - 8$$

$$= (x+1)(x^3 - 6x^2 + 10x - 8)$$

$$= (x+1)(x-4)(x^2 - 2x + 2)$$

$$\begin{array}{r|rrrr} 4 & 1 & -6 & 10 & -8 \\ & 0 & 4 & -8 & 8 \\ \hline & 1 & -2 & 2 & 0 \end{array}$$

$$\begin{aligned} \text{ii) } P(x) &= (x+1)(x-4)(x^2 - 2x + 2) \\ &= (x+1)(x-4)[(x-1)^2 - 1^2] \\ &= (x+1)(x-4)(x-1-i)(x-1+i) \end{aligned}$$

$$\text{e) } x^3 + 2x^2 + bx - 16 = 0 \quad \alpha\beta = 4$$

$$\begin{aligned} \alpha\beta\gamma &= 16 \\ 4\gamma &= 16 \\ \gamma &= 4 \end{aligned}$$

$$\begin{aligned} \alpha\beta + \alpha\gamma + \beta\gamma &= b \\ 4 + 4(\alpha + \beta) &= b \quad \text{--- ①} \\ \alpha + \beta + \gamma &= -2 \\ \alpha + \beta &= -6 \quad \text{--- ②} \end{aligned}$$

$$\begin{aligned} \text{sub @ in ① } 4 + 4(-6) &= b \\ b &= -20 \end{aligned}$$

(ii) If α, β, γ satisfy $x^3 + 2x^2 - 20x - 16 = 0$

Then $\alpha^2, \beta^2, \gamma^2$ satisfy

$$(\alpha^{\frac{1}{2}})^3 + 2(\alpha^{\frac{1}{2}})^2 - 20(\alpha^{\frac{1}{2}}) - 16 = 0$$

$$\alpha^{\frac{3}{2}} + 2\alpha - 20\alpha^{\frac{1}{2}} - 16 = 0$$

$$\alpha^{\frac{3}{2}} - 20\alpha^{\frac{1}{2}} = 16 - 2\alpha$$

$$\alpha^{\frac{1}{2}}(\alpha - 20) = 16 - 2\alpha$$

$$\alpha(\alpha - 20)^2 = (16 - 2\alpha)^2$$

$$\alpha(\alpha^2 - 40\alpha + 400) = 256 - 64\alpha + 4\alpha^2$$

$$\alpha^3 - 40\alpha^2 + 400\alpha - 4\alpha^2 + 64\alpha - 256 = 0$$

$$\alpha^3 - 44\alpha^2 + 464\alpha - 256 = 0$$

iii) $P(x) = x^3 + 2x^2 - 20x - 16 = 0$

$$\alpha^3 + 2\alpha^2 - 20\alpha - 16 = 0 \quad \text{--- (1)}$$

$$\beta^3 + 2\beta^2 - 20\beta - 16 = 0 \quad \text{--- (2)}$$

$$\gamma^3 + 2\gamma^2 - 20\gamma - 16 = 0 \quad \text{--- (3)}$$

$$\therefore \alpha^3 + \beta^3 + \gamma^3 + 2(\alpha^2 + \beta^2 + \gamma^2) - 20(\alpha + \beta + \gamma) - 48 = 0$$

$$\alpha^3 + \beta^3 + \gamma^3 + 2(44) - 20(-2) - 48 = 0$$

$$\alpha^3 + \beta^3 + \gamma^3 = -80$$

NB $\alpha^2 + \beta^2 + \gamma^2 = \frac{-b}{a}$

$$= 44 \quad \text{from part (i)}$$

$$\alpha + \beta + \gamma = \frac{-b}{a}$$

$$= 2 \quad \text{from part (i)}$$